

DLA: Mendelian Inheritance: AD and AR

PDF reading only: Most students are likely to be familiar with these basic definitions of Mendelian genetics and interpretation of pedigrees.

Review the Punnett squares and calculation of recurrence risk for AD and AR

Even if you think you know this material, it is probably a good idea to read through it anyway, please

Students must be familiar with these terms to progress in this course

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Inheritance: AD and AR – *DLAs*

SOM.MK.III.BPM1.1.FTM.3.GNET.0020	Describe basic definitions of human genetics including: locus, gene, allele, genotype, and zygosity
SOM.MK.III.BPM1.1.FTM.3.GNET.0021	From a description of a pedigree, identify autosomal dominant (AD) and autosomal recessive (AR) inheritance; Compare and contrast Mendelian and non-Mendelian genetics
SOM.MK.III.BPM1.1.FTM.3.GNET.0022	Calculate recurrence risk of autosomal dominant and autosomal recessive disorders within families
SOM.MK.III.BPM1.1.FTM.3.GNET.0023	Analyze family pedigree to determine the risk that immediate family members might be carriers of disease alleles (AR or X-linked) and when to use the 2/3 rule
SOM.MK.III.BPM1.1.FTM.3.GNET.0024	Describe and differentiate between pseudoautosomal dominance, co-dominance, and incomplete dominance; give examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0025	Describe co-dominance, giving examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0026	Describe incomplete dominance, giving examples

Inheritance: AD and AR - *Flipped classroom lecture*

SOM.MK.III.BPM1.1.FTM.3.GNET.0027	Interpret a family pedigree incorporating standard genetic symbols. Interpret a pedigree and identify features, disorders, genotype, carrier status, recurrence risk, and genetic principles
SOM.MK.III.BPM1.1.FTM.3.GNET.0028	Demonstrate how disease liability is transmitted for autosomal dominant and autosomal recessive disorders, beginning from the first "de novo" occurrence
SOM.MK.III.BPM1.1.FTM.3.GNET.0029	Propose how the basic risk calculations are performed for the different categories of genetic transmission: AR and AD
SOM.MK.III.BPM1.1.FTM.3.GNET.0030	Identify obligate carriers in a pedigree: AR, or AD with reduced penetrance
SOM.MK.III.BPM1.1.FTM.3.GNET.0031	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following autosomal dominant disorders: Huntington disease, myotonic dystrophy, familial hypercholesterolemia, osteogenesis imperfecta, Marfan syndrome, Achondroplasia, NF, acute intermittent porphyria)
SOM.MK.III.BPM1.1.FTM.3.GNET.0032	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following following autosomal recessive disorders: CF, PKU, Tay-Sachs disease, Sickle cell disease, thalassemia, hemochromatosis, homocystineuria, SCID due to ADA deficiency, alkaptonuria
SOM.MK.III.BPM1.1.FTM.3.GNET.0033	Identify pseudodominant inheritance and indicate factors that result in this inheritance pattern. Be able to identify this concept from clinical scenarios and pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0034	Compare and contrast locus heterogeneity and allelic heterogeneity; give examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0035	Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0036	Define the terms and give examples for: haploinsufficiency, dominant negative effect, gain of function mutations, loss of function mutations

Readings

- **Chapter 3 of the Korf and Irons textbook ‘Genetics and Genomics’** is focused on patterns of inheritance as it relates to human genetics
- Any student who has any issues with this material is strongly advised to read this chapter
- All students are recommended to read this chapter as much of what we discuss in the DLAs and in lecture is in the chapter

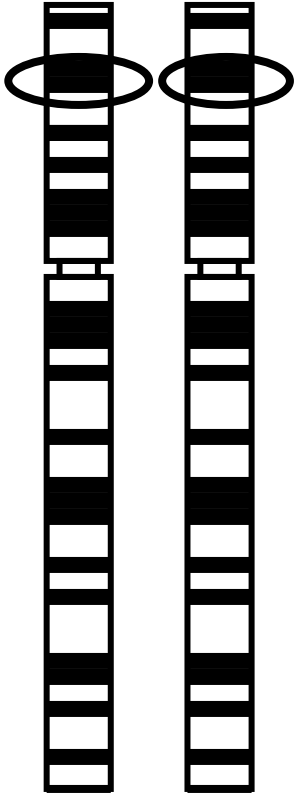
Definitions

- **Gene:** Sequence of DNA that codes for a protein and its regulatory regions
- **Locus:** Location of a gene on a chromosome
- **Allele:** Different forms (versions) of a gene
- **Polymorphism:** Multiple forms (alleles) of a gene in population (>1% of population); Note that polymorphisms are common, so a polymorphism is unlikely to be causative of a disorder.
- **Mutation (Pathogenic variant):** Change in the base sequence of the gene which results in clinical expression (phenotype). Most pathogenic variants are relatively rare in the population.
- **Homozygote:** Possess the same alleles at a locus
- **Heterozygote:** Possess different alleles at a locus
- **Dominant:** Require only one copy of the mutant allele to produce the disease phenotype
- **Recessive:** Requires two copies of the mutant allele to produce the disease phenotype

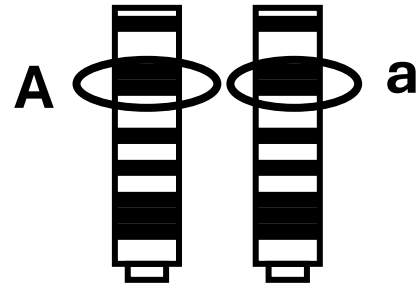
Differentiate between allele and locus

Homologous chromosomes

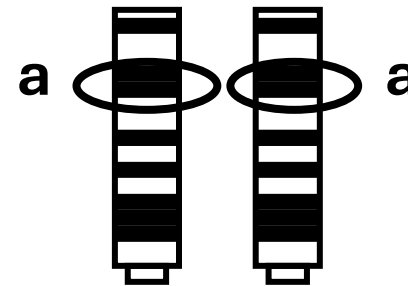
Maternal
allele A



Paternal
allele A



Heterozygote
for the gene



Homozygote for 'a'

Locus = location of a gene
(134 centimorgans or
base 60,598,996 on chromosome14p)

Allele = one of the alternative
forms of a gene that occur at
a locus (alleles A/A, A/a, a/a or
amino acid 146Phe vs146Ala)

Homozygote for
'A'

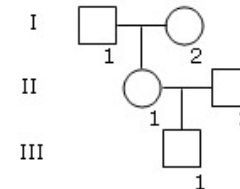
Definitions

- **Recurrence risk:** Probability that the offspring of a couple will express the genetic disorder. For single gene disorders, it does not depend on the number of previously affected/unaffected offspring. Depends on mode of inheritance of a disorder.
- **Pedigree:** The genogram or diagrammatic representation of the family history
 - Comes from the French “foot of the crane (a bird foot).
- **Consultand:** The person who approaches a physician or geneticist for a consultation.
 - This person may, or may not, be affected.
 - Consultand could be a parent of a child with a disorder.
- **Proband:** The **affected individual** in the family who gains the attention of the physician due to a genetic condition

Be familiar with pedigree nomenclature

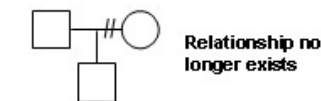
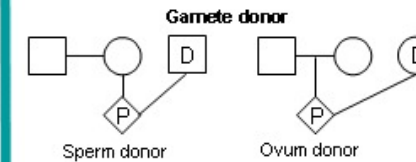
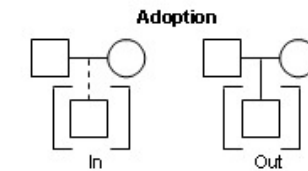
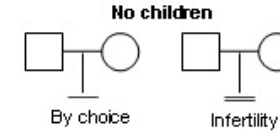
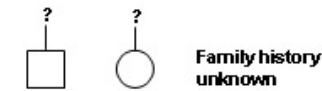
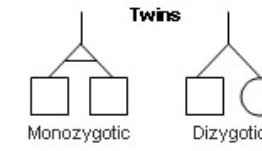
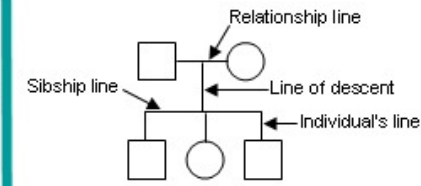
Common pedigree symbols

	Male	Female	Sex Unknown
Individual			
Affected individual			
Multiple individuals			
Multiple individuals, number unknown			
Deceased individual			
Pregnancy			
Proband			
Consultand			
Spontaneous abortion			
Termination of pregnancy			
Obligate heterozygote			
Ectopic pregnancy			

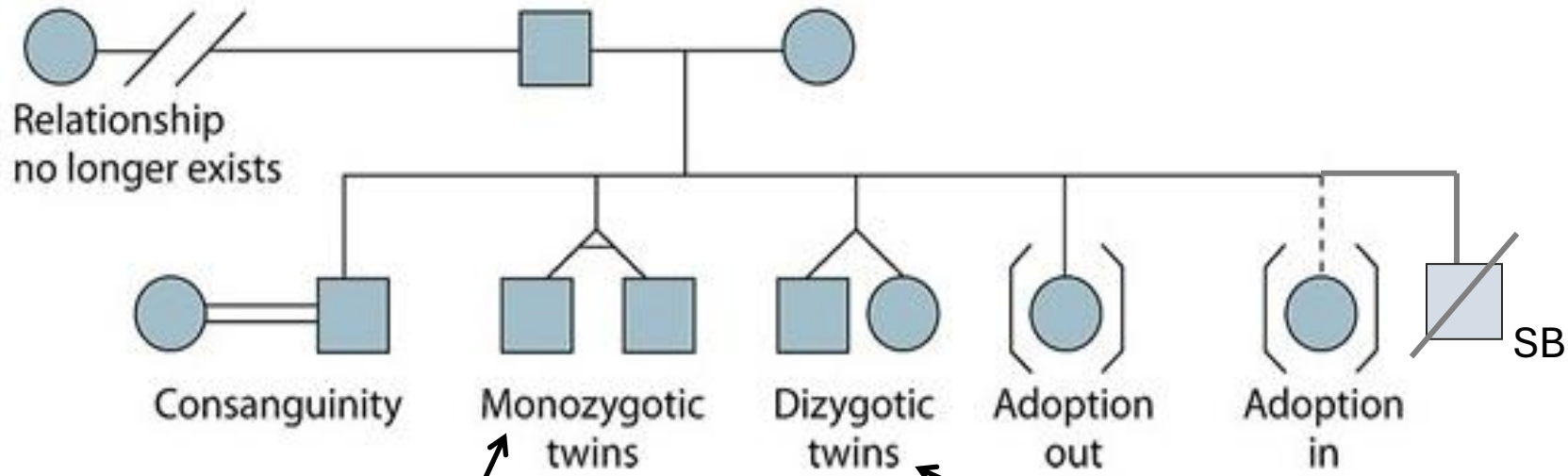


Roman numerals indicate generations; Arabic numerals indicate specific individuals within a certain generation (i.e., individual I-2 is the maternal grandmother of individual III-1).

Relationships



Pedigree chart



Arise as result of fertilization of a single zygote. Possess (just about) identical genes

Arise as result of fertilization of two ova. Genetically share about 50% genes (this is the same as any sibling)

Pedigree Nomenclature

Roman numerals

I
↓

Should recognize that one of these 'grandparents of the affected individual must be a carrier

II

Half-filled symbols may be used to indicate carriers

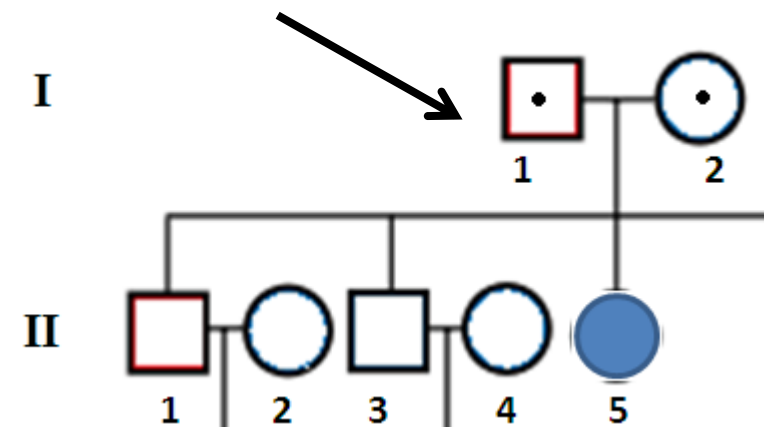
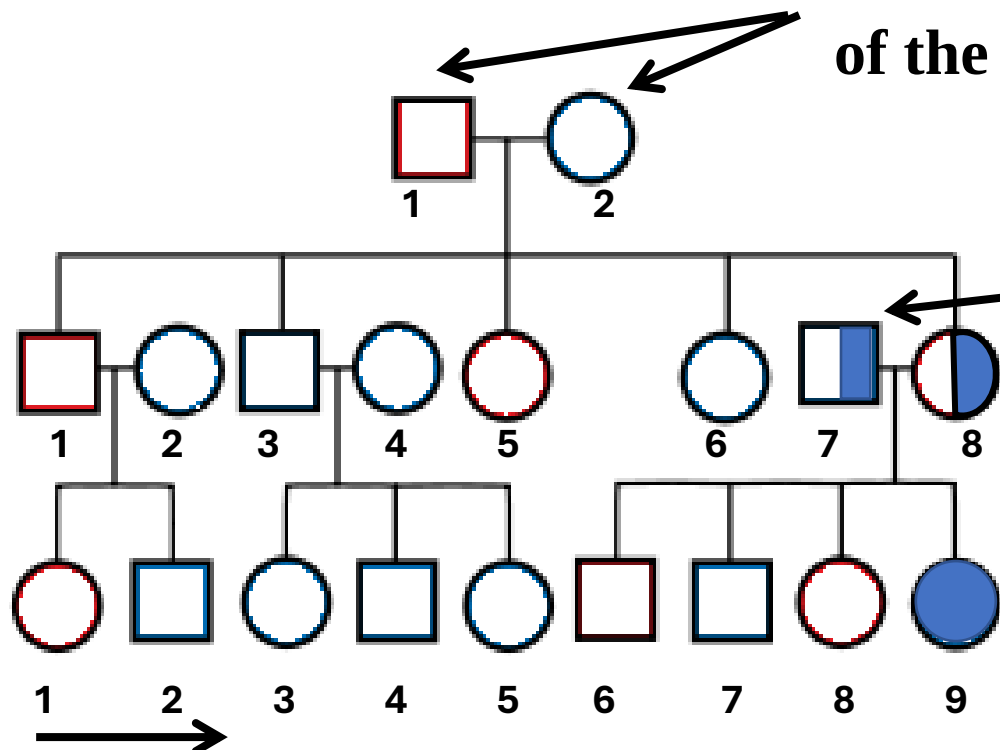
Sometimes a dot in the symbol may be used to indicate carrier status

III

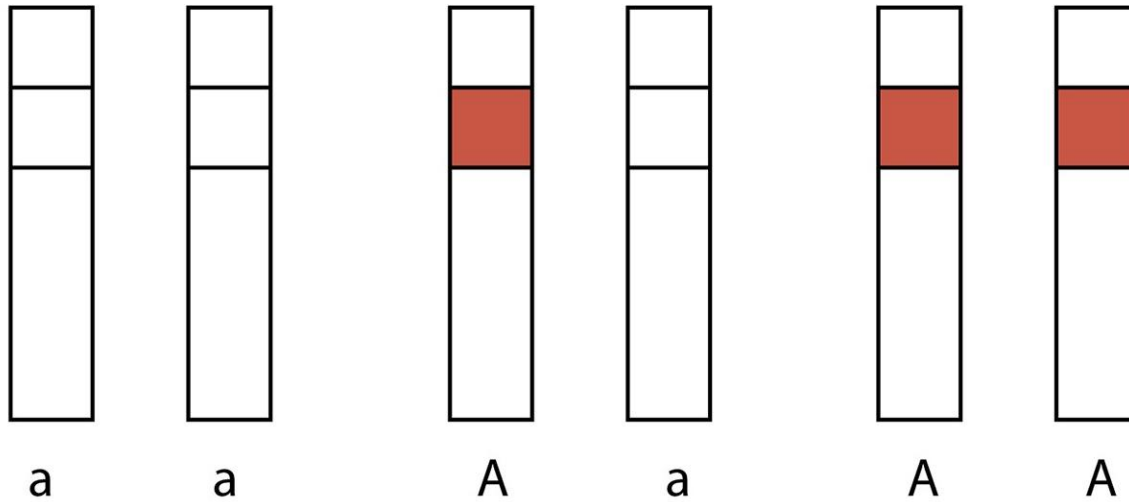
Arabic numerals

filled symbols indicate an affected individual

Recognize that the pedigrees on this slide depict an autosomal recessive trait



Autosomal Dominant disorders



Presence of one mutant allele is sufficient to cause disease phenotype

Autosomal dominant disorders manifest in the **heterozygous state**

Genotype	Homozygous	Heterozygous	Homozygous
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Phenotype	Unaffected	Affected
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Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.
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Figure 3.9 The phenotype of a dominant allele is expressed in individuals who are either homozygous or heterozygous for the allele.

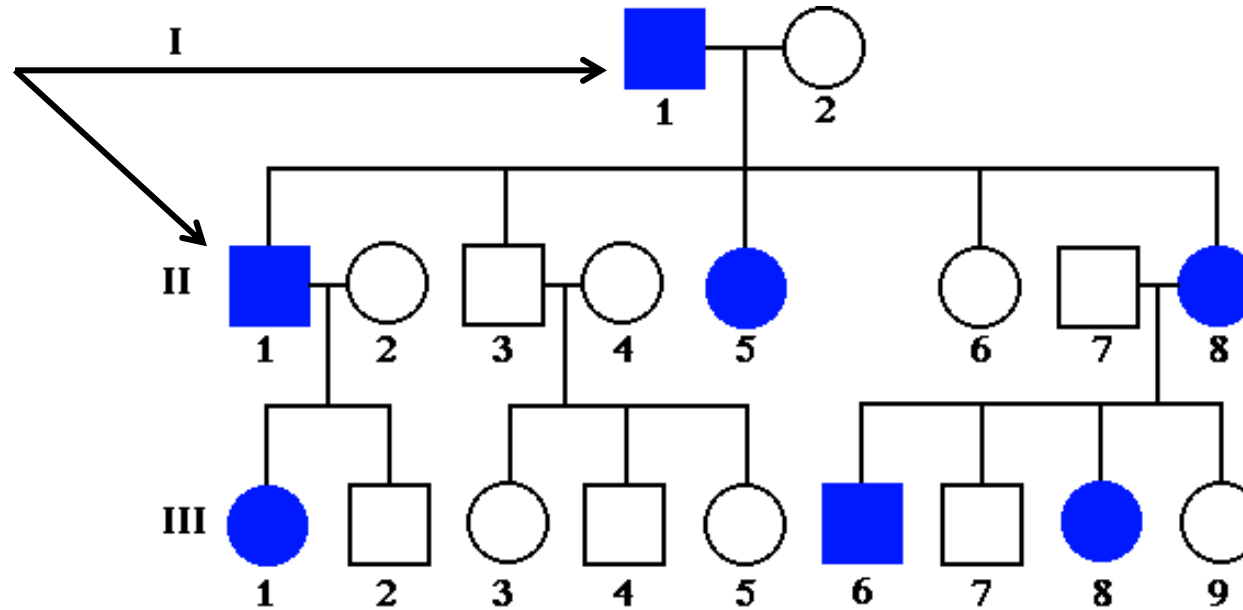
It is very rare (*in fact, it is incredibly rare*) to find an individual who is homozygous for mutations that cause autosomal dominant disorders.

It does happen, (of course), it's just that it is a very rare occurrence

One of the most common examples of a disorder in humans caused by homozygosity of dominant mutation is familial hypercholesterolemia (FHC) due to *LDLR* deficiency (note that this has symptom onset in children; 1 in 250,000 – to – 1 in a million; this is quite rare).

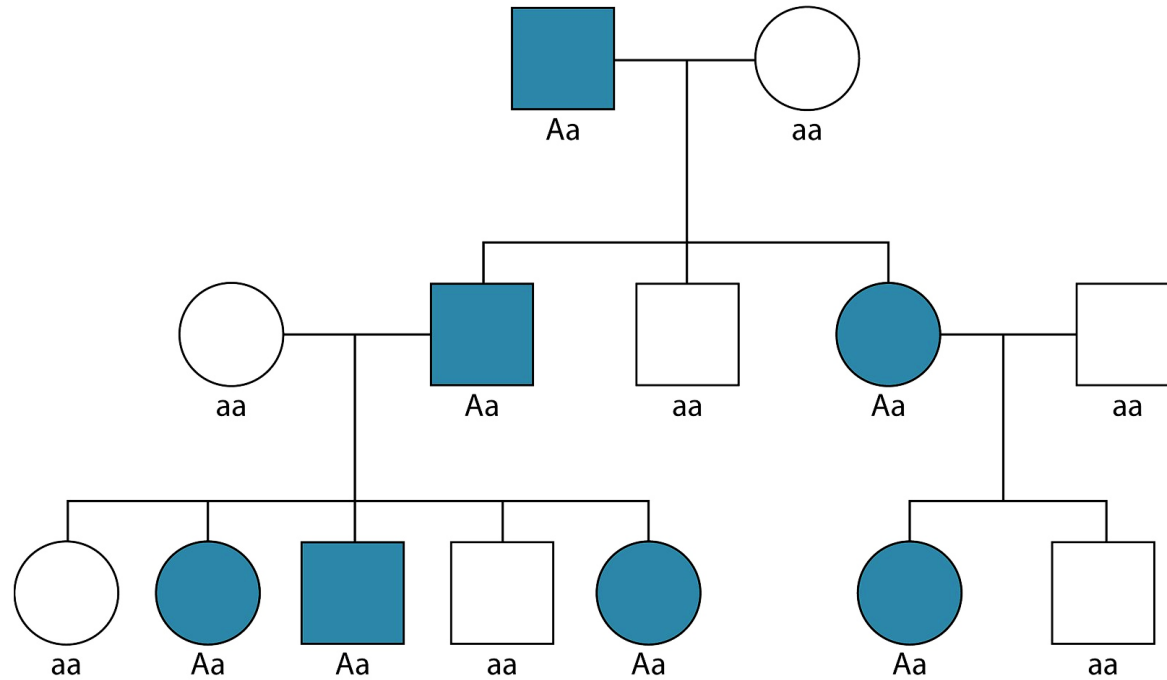
Most *LDLR* deficiency FHC is found in **heterozygotes** and is adult onset (1 in 200 – to – 1 in 500 individuals; this is of course relatively common).

Autosomal Dominant disorders



- **Affected children receive the disease-causing allele from an affected parent**
- **Skipped generations NOT common (Vertical inheritance)**
- **Males and females are typically affected with equal frequency**
- **Male to male (Father to son) transmission is seen**

Nomenclature for autosomal dominant disorders



- The disease-causing allele is dominant and is depicted as '**A**' or '**A***'
- The normal allele is recessive and is written as '**a**' or '**A**'
- The affected parent is depicted as '**Aa**' or **A*A**
- The normal parent is depicted as '**aa**' or **AA**

Recurrence Risk for Autosomal Dominant Inheritance

Normal Parent (aa) X Affected parent (Aa)

A: Pathogenic allele; a: Functional allele

		Affected parent	
		A	a
Normal parent	a	Aa	aa
	a	Aa	aa

50% risk to the next offspring regardless of sex of the child

Normal Parent (AA) X Affected parent (A*A)

A*: Pathogenic allele; A: Functional allele

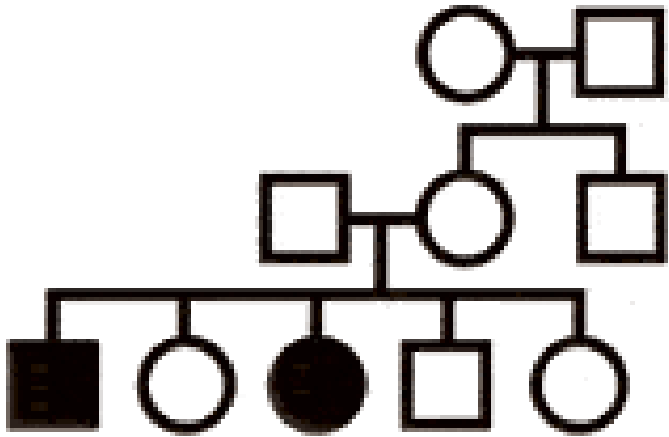
		Affected parent	
		A*	A
Normal parent	A	A*A	AA
	A	A*A	AA

50% risk to the next offspring regardless of sex of the child

Autosomal recessive (AR) disorders

- A person will show the trait (disorder) only when two disease alleles of the **same gene** are present
- AR disorders typically occur when a disease allele is inherited from both parents
- When a child has an autosomal recessive disorder, both parents are assumed to be carriers (heterozygotes) for the pathogenic allele

Autosomal Recessive disorders

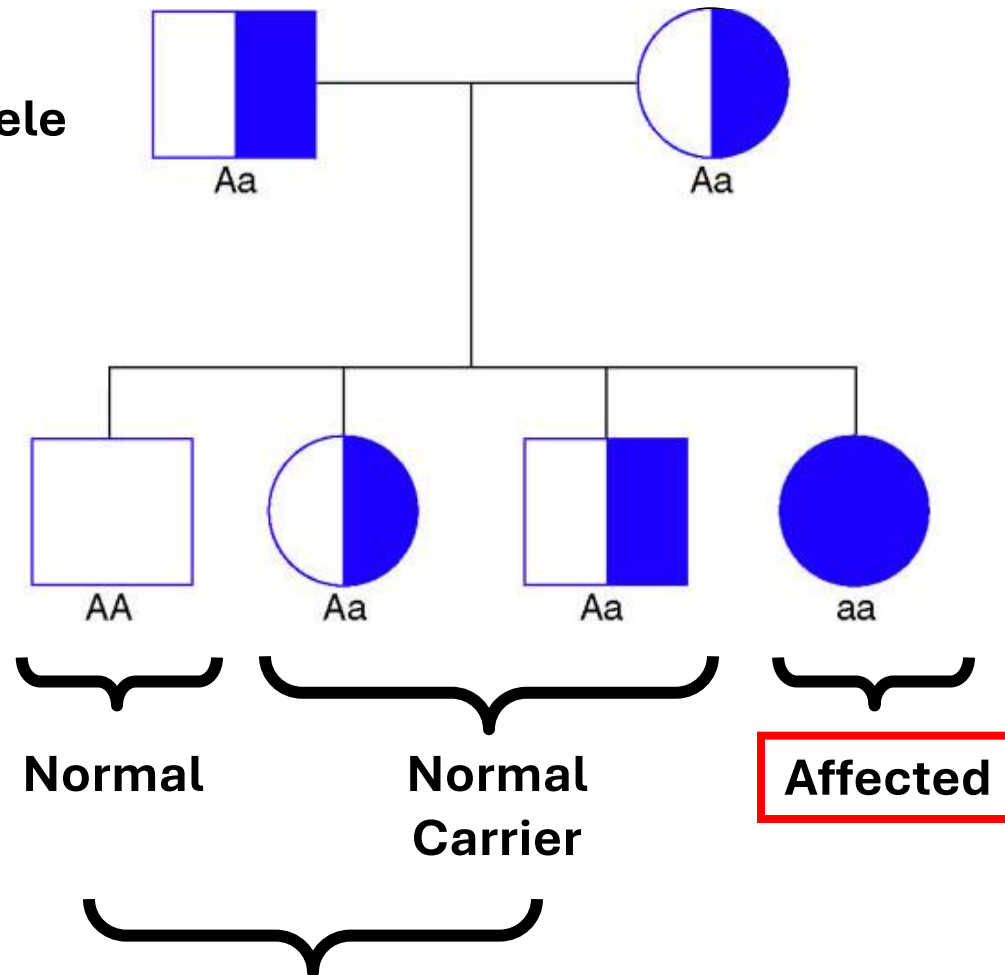


- Autosomal recessive disorders usually seen in only one generation of a pedigree (**horizontal inheritance**); Skipped generations
- Autosomal recessive disorders are expressed only in the homozygous state. Parents are usually **carriers** of the disease-causing allele. Siblings likely to be affected than others in the population (of course)
- Males and females are affected in almost equal frequency

Recurrence risk for autosomal recessive inheritance

**A: Dominant
(functional) allele**

**a: Recessive
pathogenic
variant allele**



**Each conception is an
independent event!**

**If both parents are
carriers, each product of
conception has a 25%
chance of being affected**

75% Phenotypically normal

Each conception is an independent event!

- If both parents are carriers, then: $Aa \times Aa$
 - 25% risk for the disorder with each conception if both parents are carriers
 - 25% chance that the child will inherit both functional alleles

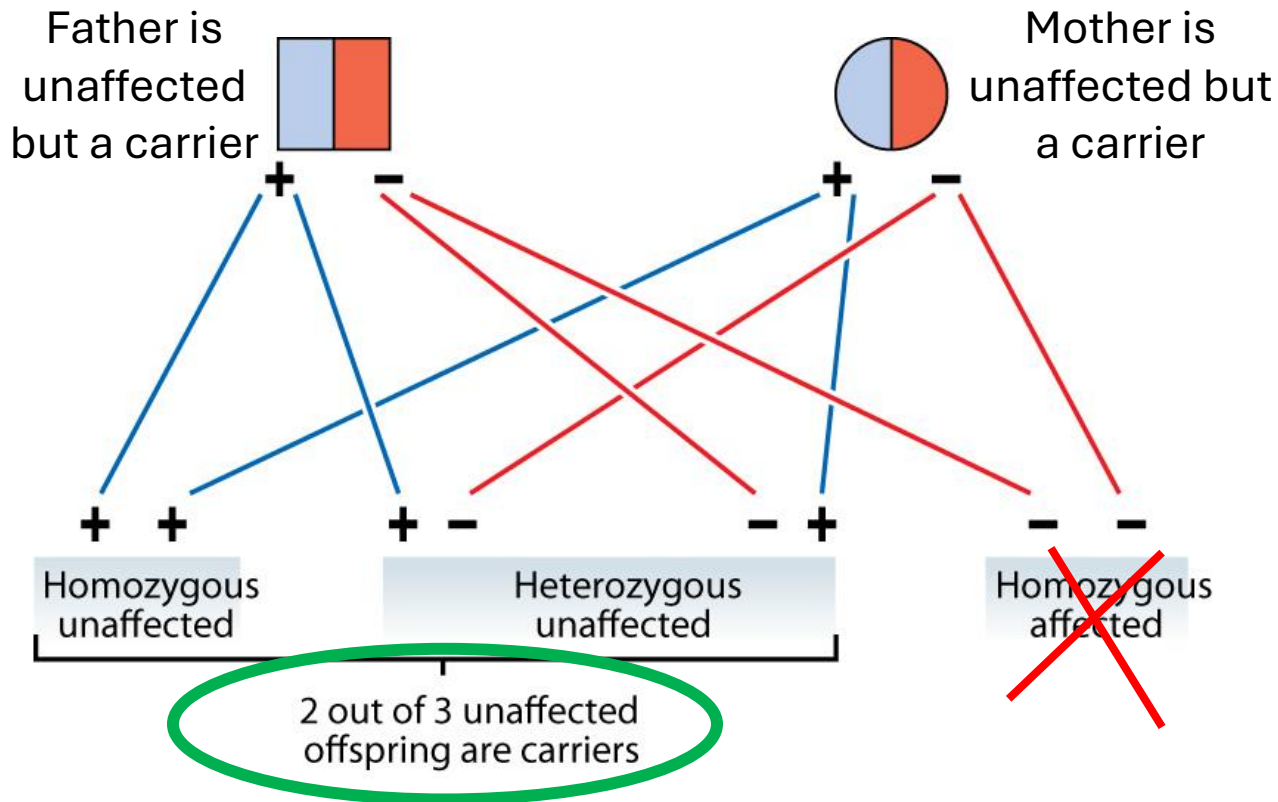
A : Dominant normal allele of the gene

a : Recessive disease-causing allele of the gene

	Carrier mother	
Carrier father	A	a
A	AA	Aa
a	Aa	aa

25%
homozygous
affected

In Autosomal Recessive Sibships (brothers and sisters) what proportion of the **Non-Affected** Sibs would be carriers?



Explanation: if sib is not affected, that sib may not be 'aa' because he/she is not affected

Therefore, there are only three other genotypes available:

AA

Aa

aA

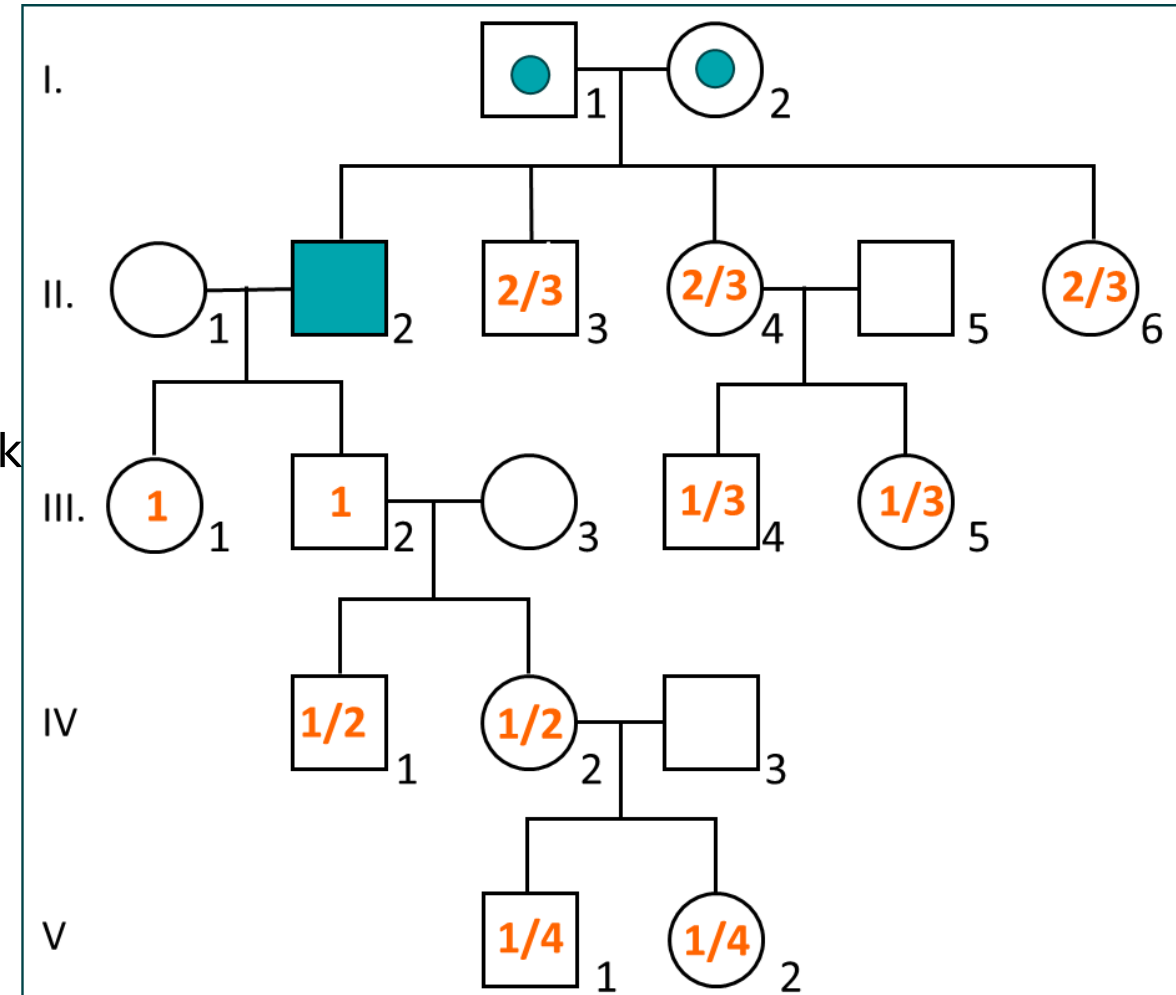
Notice that two (2) out of the three (3) are carriers:

So, for AR disorders, 2/3 of healthy sibs of a known affected person are carriers.

Caveat: disorder is highly penetrant and observable early in life.

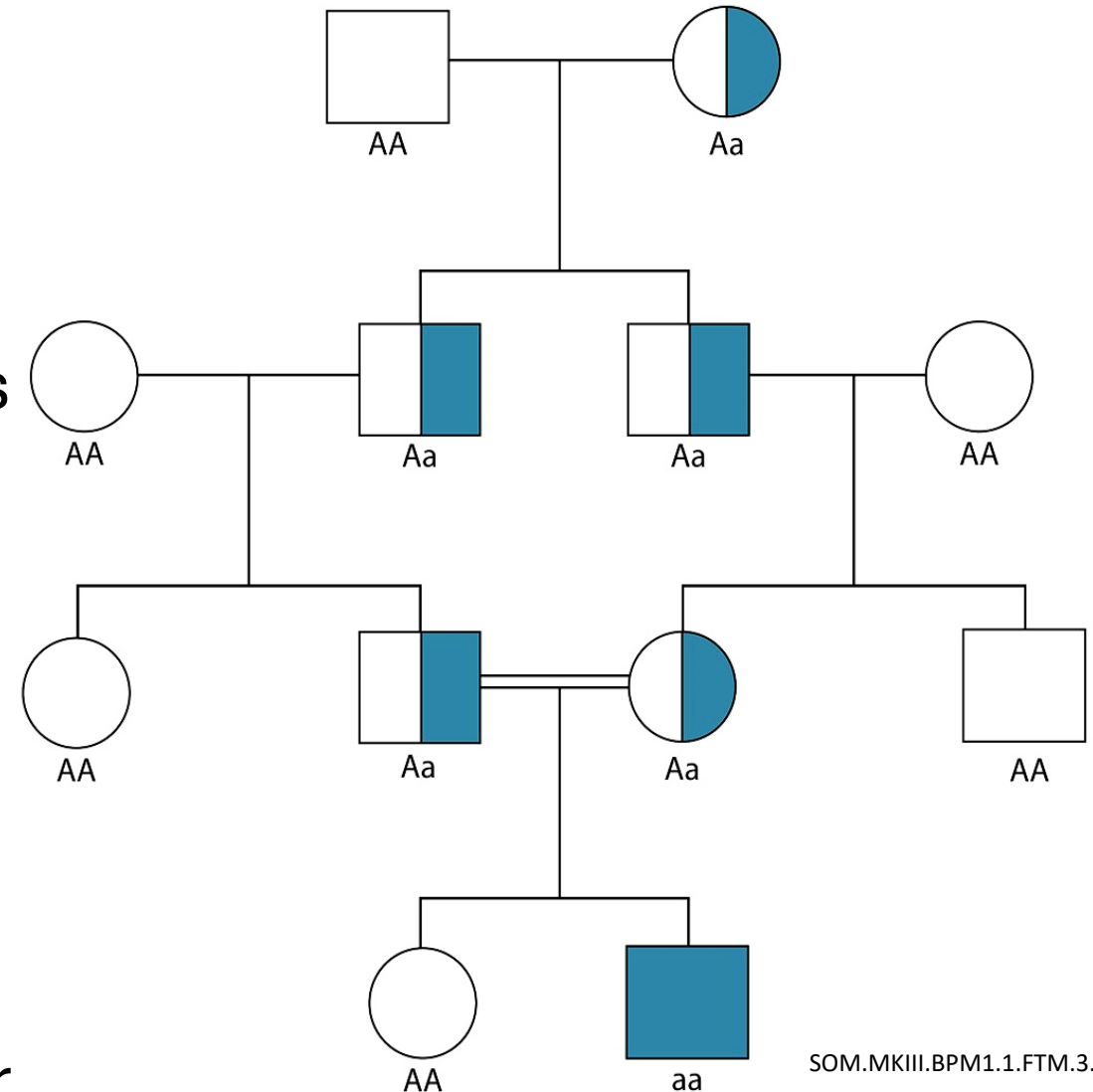
Following the risk that a person is a carrier for an allele controlling autosomal recessive traits / disorders

- The parents of an [affected](#) individual (II-2) are carriers (I-1 and I-2)
- The unaffected sibs (II-3,4,6) of an [affected](#) individual are at a $2/3$ risk of being carriers (this is the $2/3$ rule).
 - Their children would therefore be at $1/3$ risk of being carriers (III-4, 5), because they have a $1/2$ risk of passing the allele to the next generation
- The offspring of an [affected](#) individual and a non-[carrier](#) must be carriers (III-1, 2)
- Children of a [carrier](#) (III-2) and a non-carrier (III-3) are at a 50% ($1/2$) risk of being carriers (IV-1, 2).
- Children of a person at $1/2$ risk would then have a $1/4$ risk of being carriers (V-1, 2)

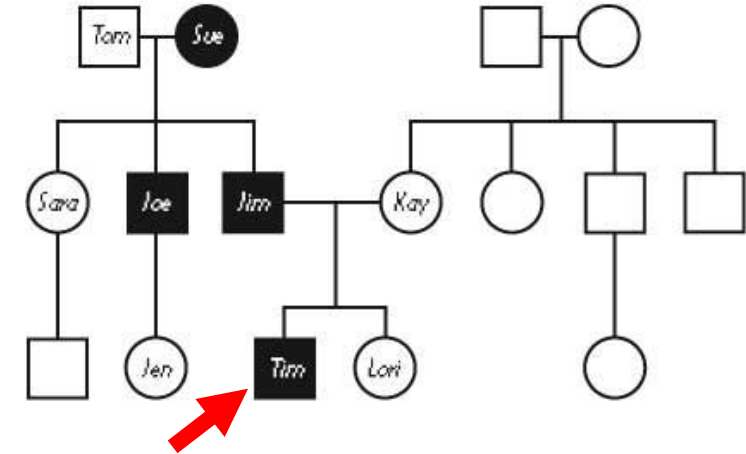


Homozygosity by descent: consanguinity increases the risk of autosomal recessive disorders

- In children born from consanguineous marriages, an allele from a common ancestor may become homozygous.
- All humans are carriers of **some** recessive mutations that if homozygous would result in severe disease, or lethality
- In this pedigree, all individuals have been genotyped for the disease associated locus (A)
- The affected individual in the 4th generation is homozygous for the allele that originated in the great grandmother



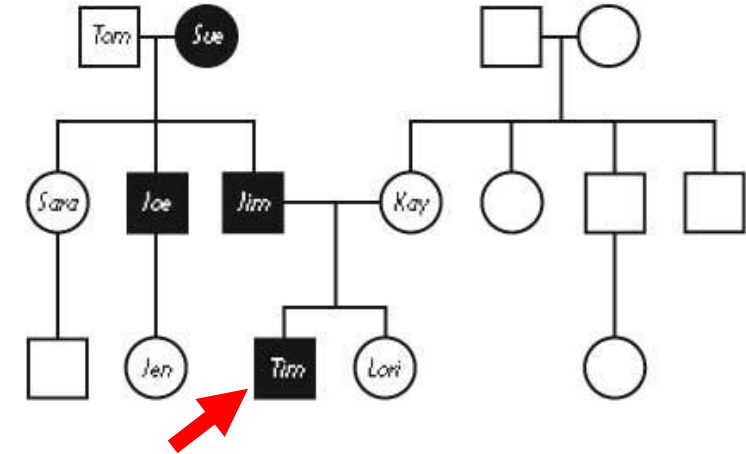
Tim (shown with the red arrow) is affected with Charcot-Marie-Tooth disease which may be inherited via many different inheritance patterns. What aspect of this inheritance pattern is suggestive of autosomal dominant inheritance?



- | | |
|---|-----|
| A. All males of an affected female are affected | 20% |
| B. The pedigree shows vertical inheritance | 20% |
| C. More males than females are affected | 20% |
| D. The disorder is in only one family line | 20% |
| E. Carrier females have affected sons | 20% |

See clinical snapshot 2.2 (p33) for a description of Charcot-Marie-Tooth disease in Korf textbook
Do not memorize the gene names for this disorder.

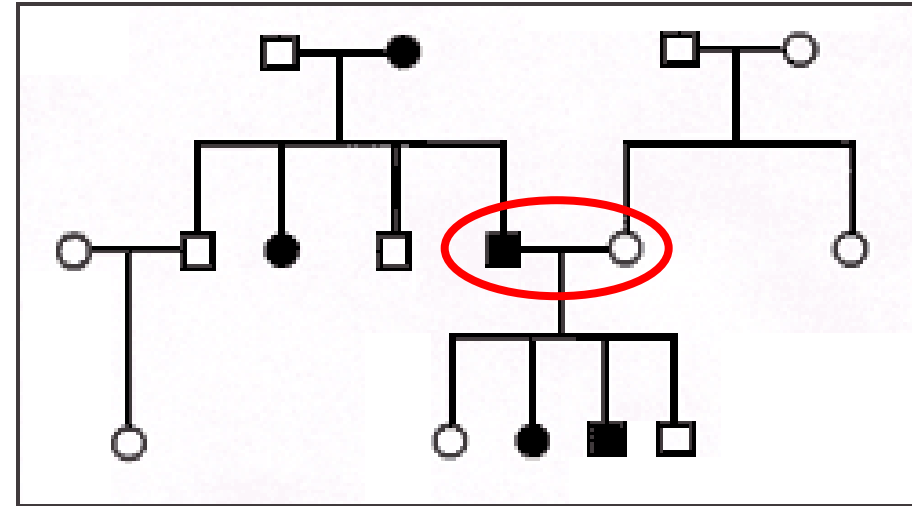
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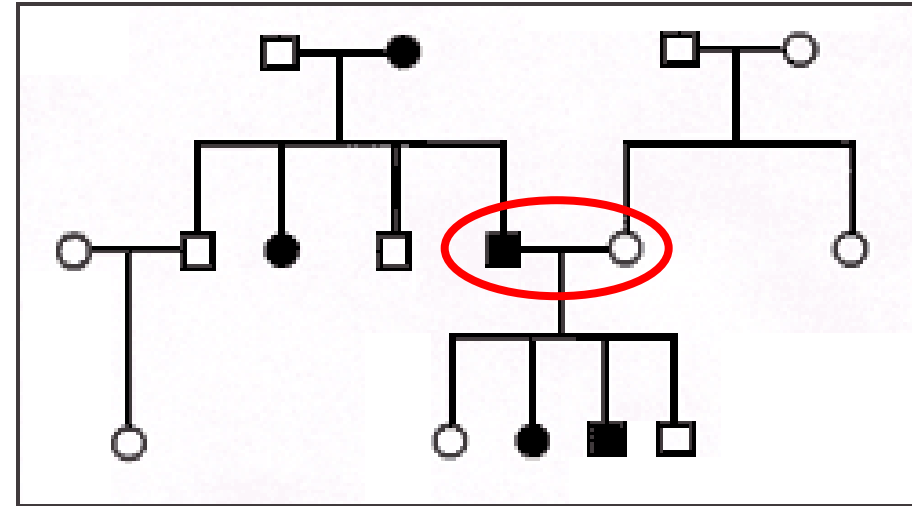
What kind of disorder is most likely shown here and what is the probability that the next child of the indicated couple will be affected?



- | | | |
|----|------|-----|
| A. | 100% | 13% |
| B. | 75% | 13% |
| C. | 67% | 13% |
| D. | 50% | 13% |
| E. | 33% | 13% |
| F. | 25% | 13% |
| G. | 5% | 13% |
| H. | Nil | 13% |

Note: Recurrence risk does not depend on the number of previously affected children. It remains the same for every new child born to the couple

What kind of disorder is most likely shown here and what is the probability that the next child of the indicated couple will be affected?



- A. 100% 13%
- B. 75% 13%
- C. 67% 13%
- 😊 D. 50% 13%
- E. 33% 13%
- F. 25% 13%
- G. 5% 13%
- H. Nil 13%

Note: for Mendelian disorders, recurrence risk does not depend on the number of previously affected children. It remains the same for every new child born to the couple; since this pedigree shows AD, each child will have a 50% risk of inheriting the disorder.

The second child of a couple has cystic fibrosis, an autosomal recessive disorder that is extremely highly penetrant, and easily detected in infancy. What is the likelihood that the first child of this couple is a carrier?

- | | | |
|----|------|-----|
| A. | 100% | 13% |
| B. | 75% | 13% |
| C. | 67% | 13% |
| D. | 50% | 13% |
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-  C. 67% 13%
- D. 50% 13%
- E. 33% 13%
- F. 25% 13%
- G. 5% 13%
- H. Nil 13%

Note: from the text, it is implied that the first child is NOT affected the first sentence says that CF is easily recognized, and always manifests. So if the first child does not have CF, then only three other genotypes are available, two of which are carriers (so $2/3$, or 66.66666%)

Reading chapter 3 in the Korf book is essential to understanding this material

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Please email if you have any questions